

Wine Score Prediction & Pricing Analysis

01. BUSINESS PROBLEM & MARKET IMPACT

BUSINESS IMPACT

Wine scores are not just ratings—they are critical economic drivers.

Expert scores directly influence pricing tiers, shelf positioning, and consumer demand forecasting.

MARKET CHALLENGE

Retailers rely on these ratings to justify premium pricing, yet score variation is complex and often disconnected from transparent metadata.

CRITICAL QUESTION

"Can we use structured metadata (Price, Winery, Region, Variety) to accurately predict and validate expert quality scores?"

02. METHODOLOGY & MODEL RESULTS



Data Preparation

Log transformation for price skewness; One-hot encoding for high-cardinality features.



Evaluation

3-fold cross-validation used for reliable performance estimation across all models.

Model Approach	MAE (Error)	R ² (Variance Explained)
Baseline (Mean Prediction)	2.48	0.00
K-Nearest Neighbors (KNN)	1.91	0.36
Random Forest Regressor	1.84	0.41

✔ Key Takeaway: The Random Forest model explains >40% of score variance using structured features alone.

03. FROM DATA TO DECISION

Feature Importance

Determining what truly drives expert scores.



CORE INSIGHT

Price is the strongest predictor of expert rating, indicating a high correlation between cost and perceived quality in the dataset.

MARKET PERCEPTION

Brand reputation (winery) significantly impacts scores. Regional effects exist but are secondary to the producer's identity.

STRATEGIC ACTION

Pricing strategy directly influences market positioning. Analytics can now inform inventory optimization and "value-for-money" selections.